Continuous Assessment Test (CAT) – I AUGUST 2025

Programme	:	B.Tech	Semester	:	Seventh
Course Code & Course Title	:	BCSE419L - Speech and Language Processing	Class Number	:	CH2025260502289 CH2025260502291 CH2025260502293
Faculty	:	Janaki Meena M Joe Dhanith P R Helen Vijitha P	Slot	:	A1 + TA1
Duration	:	90 Minutes	Max. Mark		50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information
- Use statistical tables supplied from the exam cell as necessary
- Use graph sheets supplied from the exam cell as necessary
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1		Consider a **text classification problem using multinomial naive bayes algorithm** with two classes: **C_1 = Spam, C_2 = Ham.** You are given the following training data after preprocessing **Training Documents** **Class** / **Documents**: Spam: win money now Spam: win big money Ham: meeting schedule now Ham: project meeting schedule (a) Compute the prior probabilities (2 Marks) (b) Construct the word count table for each class showing the frequency of each word in the vocabulary. (2 Marks) (c) Using Laplace smoothing, compute the conditional probabilities for all words in the vocabulary (5 Marks) (d) Classify the test document "win cash money now" by Ignoring the unknown word "cash", and using Laplace smoothing (6 Marks)	15	CO1	K3
2	a	Explain how the probability of a sentence is computed in probabilistic language modeling and how the Markov assumption helps in simplifying this computation.	6	CO1	K2

b	A unigram language model assigns the following probabilities to words: (4 Marks) **Word** | **Probability** i | 0.25 like | 0.20 deep | 0.15 learning | 0.10 NLP | 0.05 Using this model, compute the perplexity of the generated sentence: i like deep learning	4	CO1	K3
	Consider the following training sentence I like deep learning. Let context window size = 1, Initial word vectors are given below **Input (Target) Word Vectors** **Word** | **Vector** I | (0.2, 0.1) like | (0.4, 0.3) deep | (0.1, 0.5) learning | (0.6, 0.2) . List all (target, context) word pairs generated using the Skip-Gram model with window size 1 (2 marks) For the target word "like", identify: (2 marks) ➢ one positive context word, and ➢ two negative samples (k=2) Consider the below word vectors and compute the dot products between: (i) target word vector "like" and positive context word vector, and (ii) target word vector "like" and each negative sample vector. (6 marks) **Output (Context) Word Vectors** **Word** | **Vector** I | (0.1, 0.0) like | (0.3, 0.2)	10	CO2	K3

<table>
<tr><td></td><td></td><td>

deep	(0.2, 0.4)
learning	(0.5, 0.1)

</td><td></td><td></td><td></td></tr>
<tr><td>4</td><td>A</td><td>

Consider the following sentence:

Barack Obama was born in Honolulu

Assume the named entity types are:

PER – Person

LOC – Location

Format the given sentence for Named Entity Recognition (NER) using the BIO tagging scheme.

</td><td>5</td><td>CO2</td><td>K2</td></tr>
<tr><td>4</td><td>b</td><td>

Consider the following sentence:

Time flies like an arrow

An initial POS tagger assigns the following tags:

Word	Initial Tag
Time	VB
flies	NNS
like	VB
an	DT
arrow	NNS

The correct POS tags for the sentence are:

Word	Correct Tag
Time	NN
flies	VBZ
like	IN
an	DT
arrow	NN

Abbreviation	Expansion
POS	Part of Speech
NN	Noun, Singular
NNS	Noun, Plural
VB	Verb, Base Form
VBZ	Verb, 3rd Person Singular Present
IN	Preposition or

</td><td>10</td><td>CO2</td><td>K3</td></tr>
</table>

	Subordinating Conjunction
DT	Determiner

a. Identify the tagging error(s) made by the initial tagger. (2 Marks)

b. Write a Brill transformation rule that would correct the error(s) (One rule per error). (3 Marks)

c. Apply the rules step by step and show the final corrected POS tagging of the sentence. (3 Marks)

d. Briefly comment on how lexical ambiguity contributes to these tagging errors. (2 Marks)

**********All the best **************